

Helix VPN Component Library

Helix VPN — Component Library

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Overview

Enterprise-grade cross-platform UI component library for Helix VPN. All components ship with light + dark theme variants, responsive breakpoints, WCAG 2.1 AA accessibility, and platform-specific adaptations.

Platform Matrix

Component	Desktop (Tauri/React)	Mobile (Flutter)	Aurora (Qt/QML)	Web Extension
Button (5 variants)	✓	✓	✓	✓
Connection Toggle	✓ (140px)	✓ (140dp)	✓ (Cover)	✓ (mini)
Server List Item	✓ (64px)	✓ (72dp)	✓ (80px)	✓ (compact)
Protocol Badge	✓	✓	✓	✓
Latency Indicator	✓	✓	✓	✓
Input Field	✓ (48px)	✓ (56/44dp)	✓ (48px)	✓
Modal/Dialog	✓	✓	✓	—
Toast Notification	✓	✓	✓	—

Component	Desktop (Tauri/React)	Mobile (Flutter)	Aurora (Qt/QML)	Web Extension
Card	✓ (12px r)	✓ (16dp r)	✓	✓
Toggle/ Switch	✓	✓ (adaptive)	✓ (Silica)	✓
Speed Graph	✓ (80px)	✓ (80dp)	—	—
Data Usage Indicator	✓	✓	—	—
Bottom Sheet	—	✓	—	—
Tab Bar	✓	✓ (adaptive)	—	✓
Pull-to- Refresh	—	✓	✓ (Pulley)	—
Swipe Action	—	✓	✓	—
System Tray	✓	—	Cover page	Toolbar badge
Quick Settings Tile	—	✓ (Android)	—	—
Skeleton Loader	✓	✓	✓	✓
Progress Bar	✓	✓	✓	✓

Component Specifications

1. Button

Full spec: [buttons.md](#)

Variant	Background	Text	Border	Hover	Pressed	Disable
Primary	--hx-primary-500	White	None	--hx-primary-600	--hx-primary-700	38% opacity
Secondary	Transparent	--hx-primary-500	1px	overlay-hover	overlay-pressed	38% opacity
Ghost	Transparent		None			

Variant	Background	Text	Border	Hover	Pressed	Disabled
		text- primary		overlay- hover	overlay- pressed	38% opacity
Danger	--hx- semantic- error	White	None	#B71C1C	#9A0007	38% opacity
Icon	Transparent	text- primary	None	overlay- hover	overlay- pressed	38% opacity

Sizes: - Standard: 40px height, 24px padding-x, 8px radius - Compact: 32px height, 16px padding-x, 8px radius - Pill: 48px height, 32px padding-x, 9999px radius

States: Default · Hover · Pressed · Focused (2px outline) · Disabled (38% opacity) · Loading (spinner overlay)

2. Connection Toggle (Hero Component)

The signature element of Helix VPN — a large circular button driving the core action.

- **Size:** 140×140px (desktop), 140×140dp (mobile)
- **Shape:** Full circle with 4px state-colored ring
- **Icon:** Power/shield glyph, 48px white

State machine:

State	Fill	Ring	Glow	Animation
Disconnected	--hx- semantic- disconnected (#F44336)	Red	None	Static
Connecting	--hx- semantic- connecting (#FF9800)	Amber	20px blur	Spinner rotation 360°/ 1500ms
Connected	--hx- primary-500 (#00897B)	Green (#4CAF50)	20px blur, 30% opacity	Pulse scale 1.0 → 1.08, 1500ms
Error	--hx- semantic-	Red	Flash	Fast pulse 800ms

State	Fill	Ring	Glow	Animation
	error (#D32F2F)			

Transition sequence (2200ms total): 1. Tap → scale 0.95 (100ms ease-out) 2. Red → amber transition (200ms) 3. Spinner + “Connecting...” (500–2000ms, typical handshake) 4. Amber → green (300ms ease-in-out) 5. Checkmark spring-in (400ms bounce) 6. Glow pulse loop begins + stats panel slide-up (400ms)

3. Server List Item

Dimension	Desktop	Mobile	Aurora	Web
Height	64px	72dp	80px	56px
Flag icon	36×36px, 8px r	40dp circle	32px	28px
Radius	8px	—	—	6px

Composition (L → R): Flag icon → Server name + location subtitle → Latency dot + value → Protocol badge → Favorite star → Chevron

States: Default · Hover · Selected (primary tint 10%) · Pressed

4. Protocol Badge

Property	Value
Height	20px
Padding	0 6px
Bg	Primary at 10% opacity
Text	Primary, 11px/500
Radius	4px
Variants	WG, SS, MQ, OV, IK

5. Latency Indicator

Range	Color	Name	Animation
< 50ms	#4CAF50	Excellent	Static
50–100ms	#FF9800	Good	Subtle pulse

Range	Color	Name	Animation
100–200ms	#F57C00	Fair	Pulse
> 200ms	#F44336	Poor	Fast pulse

Dot size: 8px, with 4px glow (same color, 40% opacity)

6. Input Fields

Dimension	Desktop	Android	iOS	Aurora
Height	48px	56dp	44dp	48px
Radius	8px	8dp	8pt	0 (Silica rect)
Font	Body Medium	Body Medium	Body Medium	Body Medium
Focus	2px outline	2dp outline	2pt border	2px underline

States: Default · Focused (primary) · Error (red) · Disabled (38% opacity)

7. Modal / Dialog

Property	Value
Min width	360px / max 560px
Radius	16px (24px top for bottom sheets)
Shadow	Level 3
Backdrop	--hx-scrim with 150ms fade-in
Header	Title Large (22px/600) + close button
Actions	Right-aligned, primary + secondary

8. Toast Notification

Property	Value
Max width	400px
Height	48px

Property	Value
Radius	8px
Border-left	4px semantic color
Shadow	Level 2
Duration	4000ms auto-dismiss
Animation	Slide in from top-right, 300ms ease-out

9. Card

Property	Value
Bg	Surface secondary
Radius	12px (desktop) / 16dp (mobile)
Padding	16px
Shadow	Level 1 (flat preferred)

10. Toggle / Switch

Adaptive per platform: - **Android:** Material 3 Switch, Primary active track (28dp × 48dp) - **iOS:** CupertinoSwitch, Primary active color (32dp × 52dp) - **Desktop:** Custom toggle with CSS transition (200ms) - **Aurora:** Silica slider-style

11. Speed Graph / Sparkline

Property	Value
Height	80px
Download line	#00897B, 2px stroke, gradient fill
Upload line	#00BCD4, 2px dashed, gradient fill
Grid	Dotted at 25/50/75%
Update	1000ms

12. System Tray Icon

Platform	Format	Badge
macOS	Template PNG 22×22	Native overlay
Windows	.ico 16×16	Custom overlay (8px dot)
Linux	StatusNotifierItem 22×22	Native

Badge colors: green (connected), amber (connecting), red (disconnected/error)

Platform-Specific Component Collections

Desktop (Tauri/React)

- [Desktop Components](#)
- System tray context menu
- Menu bar integration
- Settings tab navigation
- Drag-and-drop server reordering
- Window management (snap, resize, minimize-to-tray)

Mobile (Flutter — Android/iOS/HarmonyOS)

- [Mobile Components](#)
- Bottom sheet (50% → 90%, 24dp top radius)
- Tab bar (Android: Material / iOS: Cupertino)
- Pull-to-refresh (64dp threshold)
- Swipe actions (72dp width)
- Quick Settings tile (Android, 1×1)
- Home screen widget (Android 4×1, iOS small/medium)
- Persistent notification with in-line controls

Aurora OS (Qt6/QML — Silica)

- [Aurora Components](#)
- Pulley menu (pull-down, 56px items)
- Cover page (1:1, gradient + status dot)
- Context menu (long-press, 48px items)
- Silica text fields (underline focus)
- Dialogs (full-width stacked actions)

- Slider (4px track, 20px thumb)

Web Extension

- Web Components
- Popup UI (380×480px max)
- Toolbar badge (state dot)
- Options page (tabbed, 600px)
- Native messaging indicator
- Content script controls